

Adversarial Review of DCA-Trie: Where the Project Is Wrong, Why It Is Wrong, and What Would Fix It

Bernard Katamanso

August 1, 2026

Abstract

This document is a deliberate, adversarial review of the DCA-Trie project. Its purpose is not to defend the work but to find where the claims, the evidence, or the evaluation are wrong. It surfaces five concrete problems: (1) the evaluation metric is lenient; (2) the empirical cost–benefit of filtering is negative; (3) Freebase ID filtering silently destroys answer reachability; (4) the ontology is both too coarse and too incomplete for the gates to be discriminating; and (5) the claims about the fixed v2 are unverified. For each we state the evidence, the mechanism, and what would fix it.

1 Problem 1: The Evaluation Metric Is Lenient

The Hits@1 metric uses substring containment rather than exact matching. In `src/utils/qa_utils.py`:

```
def match(s1, s2):  
    s1 = normalize(s1)  
    s2 = normalize(s2)  
    return s2 in s1      # answer must be a SUBSTRING of prediction
```

The prediction is scored correct if the normalized answer string appears anywhere inside it. Consequences:

- A prediction of “Jamaican” scores a hit against the ground truth “Jamaican Creole English Language.”
- Verbose or partially-correct generations are forgiven.
- The number does not measure *faithful* reasoning — the exact thing graph-constrained decoding claims to provide.

Where this is wrong: the headline numbers (89%, 88%) are upper bounds on faithful reasoning. The gap between Hits@1 and exact-match Hits@1 was never measured.

Fix: report exact-match Hits@1 alongside substring Hits@1, and report a per-path validity score (is every hop an actual KG edge?).

2 Problem 2: The Filtering Cost–Benefit Is Negative

The central claim of DCA-Trie v1 is that TypeOracle filtering “reduces paths while preserving accuracy.” The evidence says otherwise:

Dataset	Path reduction	Accuracy Δ	Time Δ
WebQSP (100)	13.3%	-1.0pp	-0.13s/q
CWQ (100)	10.4%	-4.0pp	-0.10s/q

Filtering removes paths but *also removes correct answers*, and the accuracy loss grows with graph depth (-1pp at 2 hops, -4pp at 4 hops). The time saved (~ 0.1 s/q) is negligible relative to the $6+ \text{ s/q}$ that beam search costs.

Where this is wrong: the value proposition of v1 is “cheaper for the same accuracy.” It delivers “slightly cheaper and slightly worse,” with the accuracy penalty concentrated exactly where the work claims to shine (deep graphs).

Fix: either (a) show filtering helps on a domain where the ontology is dense and correct, or (b) reframe the contribution as a *validation* layer (catch wrong predictions) rather than a *pruning* layer.

3 Problem 3: Freebase ID Filtering Destroys Reachability

The v2 fix to the “entity name vs ID” mismatch was to *filter Freebase IDs out of the trie*:

```
_FB_ID_RE = re.compile(r"^[gm]\.\w+$")
def _get_gated_paths(nx_graph, head_entity, oracle, ...):
    for neighbor in nx_graph.neighbors(head_entity):
        if _is_freebase_id(neighbor):
            continue # <-- silently drops the path
    ...
```

This “fix” introduces a *recall* bug. Measured on the 10-sample demo set:

	Answers reachable	Lost
After TypeOracle gates (v1)	94%	6%
After v2 Freebase-ID filtering	89%	11%

Eight of 72 answers (11%) become *structurally unreachable* because the only path to them passes through a Freebase ID entity such as `m.04kq1r9`.

Where this is wrong: the “fix” guarantees the model can never generate the correct answer for 11% of questions. The trie can only output names, but the KG stores many facts via IDs. Filtering the IDs does not make the model smarter — it removes the evidence.

Fix: *canonicalization*, not filtering. Map Freebase IDs to their human-readable names via the KG’s `common.topic` metadata before building the trie, so the path *name* is in the trie while the *edge* still points at the ID entity.

4 Problem 4: The Ontology Is Both Coarse and Incomplete

The TypeOracle’s discriminating power is bounded by two facts:

4.1 Coarseness

Nine hand-curated type groups and 38 relations. The ranges are broad: `people.person.nationality` has range = 7 location types. A gate that admits anything typed “Location” prunes almost nothing.

4.2 Incompleteness

In the WebQSP subgraphs, roughly 78% of nodes are Freebase IDs with *no* recorded types. Both gates are *conservative*: unknown entity $\Rightarrow$ allow. So the gates only fire on the minority of entities whose types are known, and even then only when those types conflict with a relation range.

Where this is wrong: the two sources of information (hand-curated schema, auto-mined schema) both depend on type coverage, which is sparse. The gates are permissive not by design choice but because the ontology cannot discriminate. The 13% reduction is the *ceiling* of what this ontology can prune — and it comes at a negative accuracy cost.

Fix: demonstrate the approach on a *controlled* ontology (substation IEC 61850, medical SNOMED-CT) where types are complete and ranges are narrow. The Freebase experiment can only ever show weak gates.

5 Problem 5: The Fixed v2 Claims Are Unverified

Commit b27570e claims: “v2 beam search + Freebase ID filtering + demo scripts.” The intent is “post-fix v2 should be comparable to v1.” The evidence:

- The only post-fix run is a 3-sample smoke test (33%).
- The 56%/54.9% figures predate the context-accumulation fix.
- No evaluation of beam-size > 1 , score tracking, or diversification has been run.
- The claim “comparable to v1” is *assumed*, not shown.

Where this is wrong: the project is committing to a narrative (“v2 is fixed”) before the data exists. The v2 architecture had already been declared “conclusively false” in the earlier CHAPTER4_RESULTS notes; the new fixes address symptoms (beam search, IDs, scores) but not the architectural problem (tokenization misalignment in per-hop trie construction).

Fix: run the post-fix evaluation *before* continuing to write about v2. If the fix does not recover v1-level accuracy, the correct conclusion is that iterative per-hop decoding is not viable with this tokenizer, and the thesis should pivot to the v1 + validation direction.

6 What Is Right

To be fair, the adversarial review should state what holds up:

- The **baseline reproduction** (89% vs published 91.6%) is solid.
- The **beam search finding** (+8–9pp over greedy) is a genuine, useful contribution.
- The **conceptual framing** — well-formed chains (DoG) are necessary but not sufficient; semantic gates add a real constraint — is intellectually sound and publishable as a position/analysis.
- The **domain-ontology vision** (living graphs, controlled schemas) is the correct direction and has not been tested yet.

7 Recommendation

The project’s strongest current position is not “filtering improves KGQA accuracy.” It is:

1. **Reproduce** the honest numbers: exact-match Hits@1, path-validity, and a per-error-class breakdown (type errors vs. retrieval errors vs. extraction errors).
2. **Reframe** filtering as *validation*: use TypeOracle as a post-hoc correctness check (already measured: catches 23.8% of errors at 95.5% precision), not as a pruning pre-step.
3. **Test** the controlled-ontology thesis on a domain graph, where the gates can finally be strong.
4. **Retire or rigorously re-test** v2; stop claiming fixes without post-fix evaluation.